DOMAIN
FINTECH

AI-Powered Debt Collections Platform for Banks and NBFCs

PS DEFINITION:

Recover-IQ offers a single, mobile-first platform that converts overdue-account data and field activity into actionable priorities (Risk Scoring) and coordinated recovery actions (Field and Digital Channels), offering true visibility at scale.

SHORT DESCRIPTION:

- Lenders still chase overdue accounts using spreadsheet-based allocation, delayed updates, limited visibility into field activity, inefficient field efforts, and poorly maintained data. These gaps result in slow recovery, high cost-to-collect, and compliance that is difficult to prove, while every delinquent account is treated alike.

- The RecoverIQ platform addresses these challenges through a unified, real-time approach that supports field agents, managers, and executives. By maintaining one shared case record per delinquent account, RecoverIQ ensures that an agent, a call centre, or a digital nudge can act on a case from any channel, and the case belongs to the portfolio, not to a device or an agent.

RecoverIQ integrates three core functional pillars:

- Risk-Based Prioritisation & Channel Decision: Scores each overdue account, ranks the work queue by priority, and sends high-risk cases to skilled field agents and lower-risk cases to automated digital self-cure.

- Field Enablement & Digital Self-Cure: Equips agents with a mobile-first worklist and case history, while SMS, WhatsApp, and email nudges with secure payment links resolve lower-risk cases without a visit.

- Manager & Executive Visibility: Gives managers and executives one live view of recovery, with promise-to-pay tracking, escalation, and a complete audit trail.

By unifying case data, field activity, digital engagement, and payment confirmation, RecoverIQ converts fragmented collections inputs into actionable intelligence, enabling evidence-based decisions and scalable, auditable recovery.

QUASAR 5.0

BUSINESS REQUIREMENT:

The business requirements for RecoverIQ must emphasize the combined value of integrating risk-based prioritisation, field and digital recovery, and manager-level visibility into one auditable system. The solution must be mobile-first, auditable, compliance-aware, and scalable from a branch pilot to a multi-region, multi-portfolio rollout. An AI-driven approach is encouraged, but rule-based logic is equally acceptable.

CORE FEATURES:

- Case Intake: The platform must load overdue accounts, including how long each has been overdue, into the system and create exactly one case per account without duplicates.
- Risk Scoring: The system must calculate a dynamic score for every case, with reasons, using rule-based logic or AI, and rank the work queue by priority.
- Channel Decision: The system must route high-risk cases to a field agent and low-risk cases to the digital self-cure track.
- Allocation: The system must assign cases to agents based on agent proximity, performance, and risk, with support for reassignment and workload balancing.

Field & Digital Engagement

- Agent Mobile App: The application must give agents a prioritised worklist, full case history, and call and visit logs, with safe offline synchronisation for weak connectivity.
- Digital Nudges: The platform must send SMS, WhatsApp, and email nudges with secure payment links while enforcing frequency caps and quiet hours.
- Payments: The system must support on-the-spot payment capture or payment links and mark a case as paid only after confirmation from the payment gateway, preventing duplicate posting.
- Promise-to-Pay + Escalation: The system must track every promise to pay, automatically flag broken promises with the next action or manager escalation, and route disputes to the appropriate owner.

Visibility, Motivation & Learning

- Dashboards: The system must provide role-restricted dashboards showing collection rate, ageing, cost per recovery, agent performance, and live field activity.
- Agent Motivation: The platform must offer a leaderboard and progress view for agents, with no fabricated rankings.
- Learning Loop: The system must feed outcomes back into risk scores and next-best actions while remaining explainable and subject to manager review.

Compliance, Security & Audit

- Compliance + Audit: The platform must log every action and enforce contact-time and fair-practice rules.
- Security + Privacy: The platform must apply least-privilege role-based access, encryption, masked customer data, and a data-retention policy.

Streamlined Recovery Execution

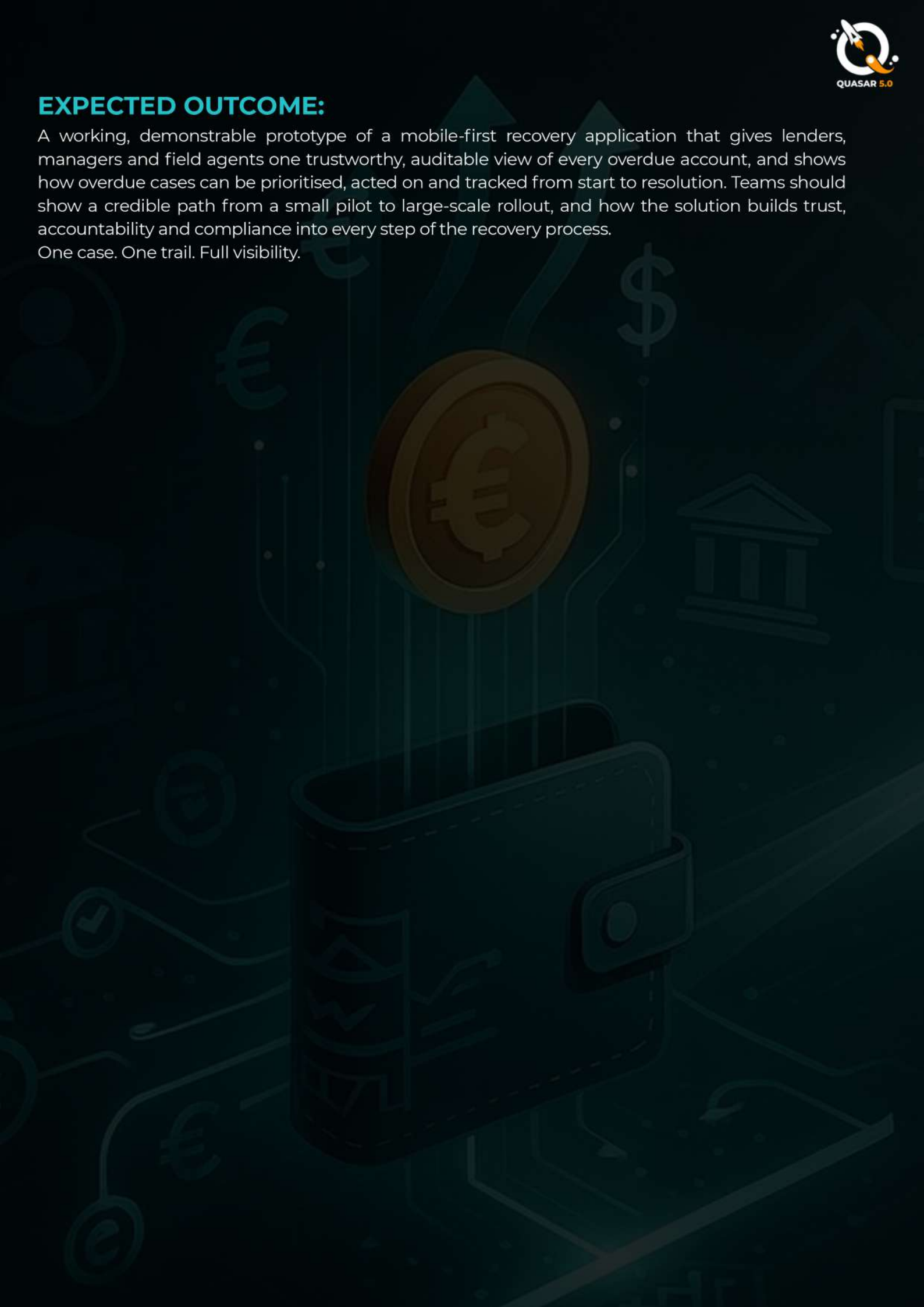
- The application must maintain a single shared case record that both agents and managers can retrieve from separate sessions, with consistent status across agent, manager, and executive views. A case must never be shown as paid or closed until the payment system confirms the transaction.

QUASAR 5.0

EXPECTED OUTCOME:

A working, demonstrable prototype of a mobile-first recovery application that gives lenders, managers and field agents one trustworthy, auditable view of every overdue account, and shows how overdue cases can be prioritised, acted on and tracked from start to resolution. Teams should show a credible path from a small pilot to large-scale rollout, and how the solution builds trust, accountability and compliance into every step of the recovery process.

One case. One trail. Full visibility.



PS :- FINTECH ESSENTIAL REQUIREMENTS

QUASAR 5.0

Essential Requirements	Description
Case Intake	Overdue accounts loaded into the system; exactly one case per account, no duplicates.
Risk Scoring	Every case gets a score with reasons, using rule-based logic or AI, and the queue is ranked by priority.
Channel Decision	High-risk cases go to field agents; low-risk cases go to the digital self-care track.
Case Allocation	Cases allocated to agents by pin code and DPD (days past due) bucket, with reassignment and workload balance.
Agent Mobile App	Mobile-first app with a prioritised worklist, full case history, and a call and visit log.
Visit & Location Tracking	Each visit is stamped with time and GPS location at check-in, so managers can verify field activity.
Digital Nudges	SMS, WhatsApp and email reminders with secure payment links, plus frequency caps and quiet hours.
Payment Collection	On-spot capture or secure payment link, with an instant digital receipt to the borrower.
Payment Confirmation	A case is marked paid only after the gateway confirms, with no double posting.
Promise-to-Pay & Escalation	Every promise tracked; a broken promise auto-flags the next action or the manager, and overdue promises are routed.
Manager & Executive Dashboards	Live view of collection rate, ageing, cost per recovery, agent performance and field activity.
Offline Capability	Works with weak connectivity; data is cached and syncs safely when the signal returns.
Simple Permissions	Role-based views (Agent vs Level-2 Manager vs Level-1 Manager vs Executive) with masked customer data to prevent unauthorised access.
Audit & Compliance	Every action logged, with contact-time and fair-practice rules enforced.
Default Classification	A borrower with more amount of missed EMIs is flagged as a default, and the rule is configurable.

PS :- FINTECH ESSENTIAL REQUIREMENTS

QUASAR 5.0

Essential Requirements	Description
City & State Sentiment View	Sentiment is aggregated by city and by state as well the individual person, showing whether the full state's repayment mood is positive or negative.
Borrower Note Capture	The agent records the borrower's actual statement (reason for non-payment, promised date, circumstances) in the borrower's own words, by typing or by voice. Speech-to-text works in multiple local languages, and the note is saved to the case with a time stamp.

DOMAIN
WASTE MANAGEMENT

AI-Powered IoT-Enabled Smart Dustbin & Decentralized Organic Waste-to-Compost Platform

PS DEFINITION:

Housing societies generate large amounts of biodegradable waste, such as fruit and vegetable peels, tea leaves, and food scraps, but its management often lacks proper segregation, automated stirring, moisture control, real-time monitoring, batch traceability, and compost maturity assessment. CompostIQ addresses these challenges through an IoT-enabled smart dustbin and composting unit connected to a centralized software platform. It collects, segregates, and weighs organic waste while using sensors for real-time monitoring, automated stirring, and controlled moisture correction. AI-assisted compost maturity estimation provides explainable recommendations and identifies when compost is ready. Each batch receives a unique identity to track its complete journey from waste intake to final harvest. The platform also manages compost inventory, quality inspection, internal reuse, and external sales through a marketplace connecting producers with potential buyers. The solution must demonstrate a functional physical prototype with actual sensor data, safe hardware operation, intelligent AI recommendations, and a scalable architecture for deployment across multiple housing societies.

SHORT DESCRIPTION:

CompostIQ is an AI-powered, IoT-enabled decentralized waste-to-compost platform that connects a physical smart dustbin installed in a housing society with automated composting operations and a centralized digital dashboard. The ESP32 collects real sensor readings from a waterproof DS18B20 temperature sensor, capacitive moisture sensor, and load cell with HX711. The system monitors compost conditions, schedules or automates stirring, recommends controlled water spraying or dry-material addition, and reassesses moisture after corrective actions. AI-assisted analytics estimate compost maturity, identify abnormal conditions, and recommend corrective actions. Every batch is assigned a unique ID and tracked through Active, Curing, and Ready for Harvest stages, with final inspection and operator approval before harvesting. Finished compost is weighed, recorded in inventory, and made available for internal reuse or external sale through a circular marketplace. The platform provides role-based access, operational alerts, batch traceability, and environmental impact reporting. One housing society. One physical smart dustbin. Automated composting. Real sensor data. Traceable batches. A circular compost marketplace.

CORE FEATURES:

1. Smart Dustbin & Organic Waste Collection: Develop and install a physical smart dustbin and composting unit in a housing society. Collect suitable organic waste, including fruit peels, vegetable peels, and tea leaves, while removing plastic, glass, metal, and other unsuitable materials.

2. IoT-Based Monitoring & Weight Tracking: Integrate an ESP32, waterproof DS18B20 temperature sensor, capacitive moisture sensor, and load cell with HX711. Monitor incoming waste weight, compost temperature, moisture, and finished compost weight using actual sensor readings.

3. Automated Stirring & Intelligent Moisture Management: Implement a motorized stirring mechanism that operates at suitable intervals. Alert operators when stirring is due and support controlled water spraying for dry compost or recommendations to add dry leaves and other suitable dry material when compost is too wet. Recheck moisture after mixing.

4. AI-Assisted Composting & Harvest Management: Analyze sensor trends and processing history to estimate compost maturity, detect abnormal conditions, and recommend corrective actions. Track batches through Active, Curing, and Ready for Harvest stages, requiring final inspection and operator approval before harvest.

5. Batch Traceability & Quality Management: Assign a unique ID to every batch and record its source, waste type, starting weight, sensor readings, stirring events, moisture corrections, processing history, inspection results, approval status, and final compost weight.

6. Compost Inventory & Circular Marketplace: Record packaged compost, manage inventory, list eligible batches, recommend potential buyers and indicative prices, and track orders, delivery, internal reuse, and external sales.

7. Unified Dashboard, Security & Impact Reporting: Provide live sensor readings, historical trends, batch status, alerts, and operational analytics. Implement role-based access for society administrators, processing operators, buyers, and platform administrators. Report waste processed, compost produced, and compost reused or sold.

BUSINESS REQUIREMENTS:

The business requirements for CompostIQ must emphasize the integration of physical smart-dustbin development, IoT-enabled monitoring, automated composting, intelligent moisture management, AI-assisted process intelligence, batch-level traceability, and circular compost commerce into a unified system.

Smart Dustbin & Organic Waste Collection

- Physical Smart Dustbin Development: The solution must include the design, development, and deployment of a functional physical smart dustbin and composting unit situated within a housing society. The unit must support suitable organic waste collection and provide an integrated mechanism for monitoring and processing the collected waste.

- Waste Collection & Segregation: The system must support separate collection and storage of wet waste and dry waste in designated bins. Suitable organic waste, including fruit peels, vegetable peels, and tea leaves, must be collected for composting. Plastic, glass, metal, and other unsuitable contaminants must be kept separate and excluded from the composting process.

- Batch Management: Every waste batch must receive a unique ID. The system must record waste type, starting weight, collection date, source, and subsequent processing information.
- Incoming Waste Weight Measurement: A load cell with an HX711 amplifier must measure incoming waste weight and associate the measurement with the correct batch and timestamp.

IoT Sensor Monitoring & Connectivity

- A waterproof DS18B20 sensor monitors compost temperature and records live and historical readings, while a capacitive moisture sensor detects whether compost is too wet or dry using calibrated thresholds.
- The ESP32 collects actual sensor data and transmits timestamped readings to the backend through Wi-Fi using MQTT or HTTP.
- The dashboard displays live readings, historical trends, batch status, and connectivity while detecting disconnected devices, outdated readings, sensor faults, and abnormal measurements.

Automated Stirring & Moisture Management

- The unit uses a motorized mechanism to stir compost at suitable intervals, improving airflow and uniform decomposition. The system tracks stirring schedules, alerts operators when stirring is due, and automatically operates the mechanism according to configured intervals and safety conditions.
- When compost is too dry, it recommends the required water quantity for controlled spraying; when too wet, it suggests adding dry leaves or other dry materials and improving airflow. A controlled water-spraying mechanism with operating limits prevents overwatering.
- After stirring, spraying, or adding dry materials, the system rechecks moisture levels to evaluate improvement, following calibrated measurements or documented operating rules.
- Timely alerts and corrective recommendations help prevent excess moisture and bad odour, while motorized stirring and spraying include proper safety interlocks and control limits.

Compost Lifecycle & Harvest Management

- Compost Stage Tracking: Each batch must have one of three lifecycle statuses: Active, Curing, or Ready for Harvest. Stage transitions must be recorded with timestamps and supporting information.
- AI-Assisted Maturity Estimation: The platform must estimate compost maturity using temperature and moisture trends, processing duration, and batch history where available. Predictions must be clearly identified as estimates.
- Batch Management: Every waste batch must receive a unique ID. The system must record waste type, starting weight, collection date, source, and subsequent processing information.
- Incoming Waste Weight Measurement: A load cell with an HX711 amplifier must measure incoming waste weight and associate the measurement with the correct batch and timestamp.

AI-ASSISTED INTELLIGENCE & CORRECTIVE ACTIONS:

- Anomaly Detection: The platform must identify unusual temperature, moisture, or weight readings using configured thresholds and AI-assisted analysis where sufficient data is available.
- Actionable Recommendations: The system must recommend relevant actions, including stirring compost, spraying controlled quantities of water, adding dry material, improving airflow, or requesting operator inspection.
- Compost Output Prediction: The platform should estimate expected compost output using incoming waste weight, processing history, and available historical data. Predicted output must be distinguished from actual measured weight.
- Explainable AI Outputs: Recommendations and maturity estimates must include understandable reasons and supporting data. The system must not present predicted measurements or maturity estimates as verified physical results.

BATCH TRACEABILITY & QUALITY VALIDATION:

Complete Batch History: Each batch must maintain a traceable history covering its source, initial weight, sensor readings, stirring events, moisture corrections, processing stages, and final compost weight.

- Processing Event Logging: The system must record relevant events, including operator actions, automated stirring, water spraying, dry-material additions, alerts, status changes, and inspections.
- Quality Checks & Operator Approval: The application must support inspection records and operator approval before a batch is marked ready for harvest or made available for marketplace listing.
- Auditability: Authorized users must be able to review batch processing and validation records. Any quality or certification claims must be supported by appropriate evidence.

COMPOST INVENTORY & MARKETPLACE:

1. Inventory Management:

The platform must record finished compost quantities, packaging information, batch IDs, and available inventory. Inventory must be updated after packaging, reuse, sale, or dispatch.

2. Compost Listings:

Authorized users must be able to list eligible compost batches with available quantity, relevant quality information, maturity status, and indicative pricing.

3. Buyer Matching & Price Recommendations:

The system should recommend potential buyers and indicative prices based on available quantity, location, demand, and recorded quality information.

4. Orders & Circular Reuse:

The platform must support order creation, inventory allocation, order status tracking, delivery coordination, and recording of compost reused within the housing society.

QUASAR 5.0

DASHBOARD, ACCESS CONTROL & IMPACT REPORTING:

1. Unified Operational Dashboard:

The dashboard must display live sensor readings, historical trends, device connectivity, batch stages, stirring schedules, moisture alerts, corrective actions, harvest readiness, inventory, and marketplace activity.

2. Role-Based Access Control:

The platform must support appropriate permissions for society administrators, processing operators, buyers, and platform administrators. Users must only access functions authorized for their roles.

3. Environmental Impact Tracking:

The system must report total waste processed, finished compost produced, and compost reused or sold. Any estimated waste diversion or environmental impact must be identified as an estimate and supported by recorded data.

4. Scalability:

The architecture must support an initial pilot with one physical smart dustbin and composting unit in one housing society, with a clear path toward managing multiple units and housing societies.

EXPECTED OUTCOME:

A demonstrable AI-powered, IoT-enabled smart-dustbin and automated composting prototype that combines physical waste collection, real-time sensor monitoring, automated stirring, intelligent moisture management, AI-assisted compost maturity estimation, batch-level traceability, and a circular compost marketplace.

The solution will enable a housing society to monitor organic waste from collection through processing and final harvest, maintain transparent batch records, identify abnormal composting conditions, and connect finished compost with potential buyers or internal users.

The prototype will establish a practical foundation for scaling decentralized organic waste management across multiple housing societies.

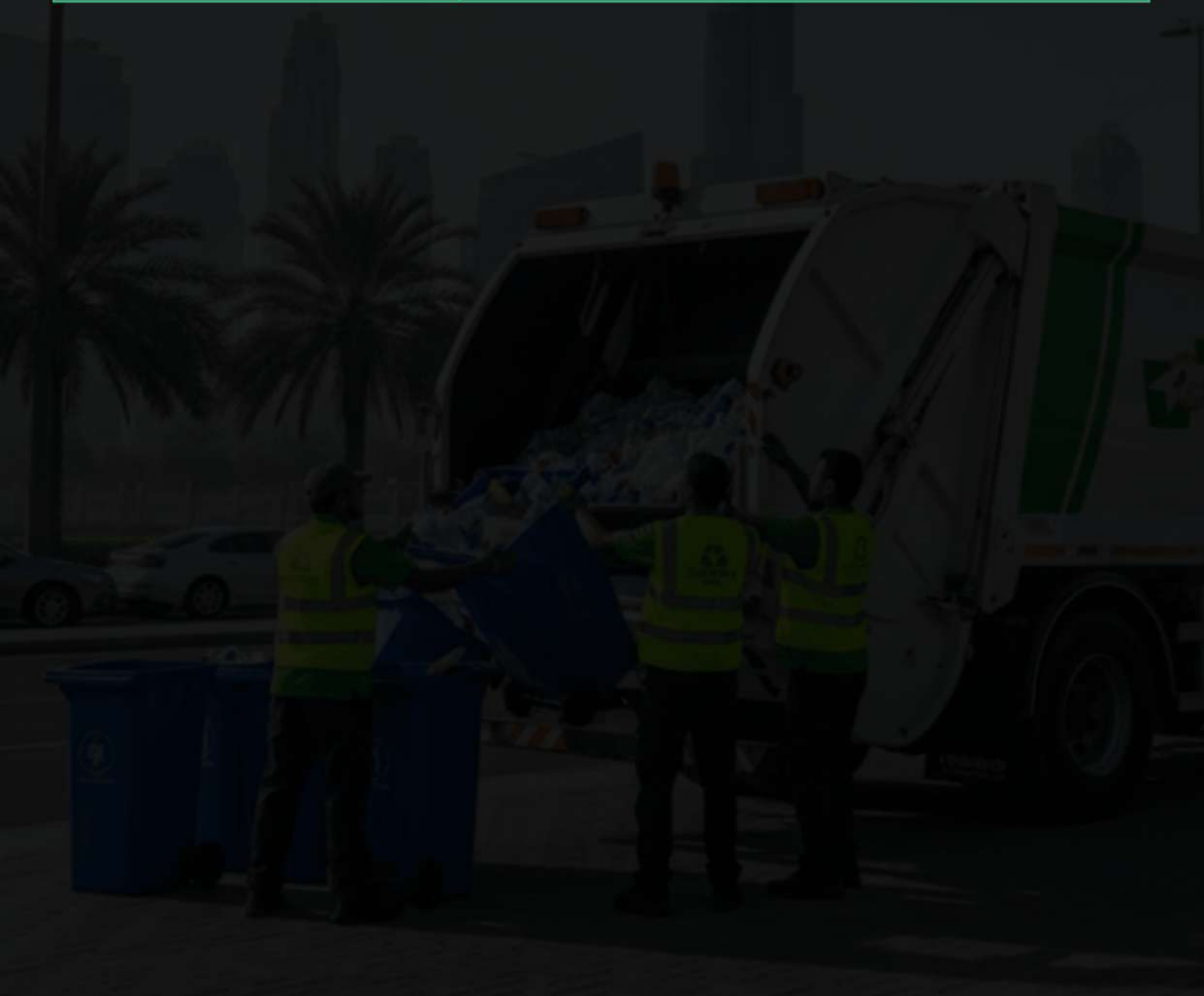
One housing society. One physical smart dustbin. Automated stirring. Intelligent moisture control. Real sensor data. Traceable compost batches. A circular compost marketplace.

PS :- WASTE MANGEMENT ESSENTIAL REQUIREMENTS

Essential Requirement	Description
Waste Segregation	Keep wet waste and dry waste separately in designated bins for proper waste collection and disposal.
Batch Management	Give each batch a unique ID. Save the waste type, starting weight, date and source.
Temperature Monitoring	A waterproof DS18B20 sensor checks the compost temperature during processing.
Moisture Monitoring	A capacitive sensor checks whether the compost is too wet or too dry and helps guide corrections.
Moisture Management & Corrections	If the compost is too dry, use a small, controlled water spray. If it is too wet, add dry leaves or other dry material and improve airflow. Check the sensor reading again.
Automatic Stirring	A motorized mechanism automatically stirs the compost at suitable intervals to improve airflow and ensure uniform decomposition.
Smart Stirring & Moisture Control	The system alerts the operator when it is time to stir the compost. It monitors moisture levels and recommends the quantity of water to spray when the waste is too dry or the quantity of dry leaves or other dry material to add when the waste is too wet. After mixing, it checks the moisture level again to maintain proper composting conditions and prevent excess moisture and bad odour.
Removable Collection Tray	Use a removable tray or drawer to take out finished compost after curing and final readiness checks.
Weight Tracking	A load cell with an HX711 module records incoming waste weight and finished compost weight.
ESP32 & Live Data	The ESP32 collects real sensor readings and sends them to the server over Wi-Fi using MQTT or HTTP.
Dashboard, Alerts & AI	Show live readings, trends and batch status. Warn the operator about unusual readings and suggest simple actions.
Show live readings, trends and batch status. Warn the operator about unusual readings and suggest simple actions.	Show Active, Curing or Ready for Harvest. Use sensor trends to estimate maturity, then check the compost before removing it.

PS :- WASTE MANGEMENT ESSENTIAL REQUIREMENTS

Essential Requirement	Description
Waste Collection & SegregationBatch History & Quality Check	Collect fruit peels, vegetable peels, tea leaves and other suitable food waste. Remove plastic, glass and metal.Save sensor readings, process steps, inspection results and operator approval for every batch.
Batch ManagementCompost Inventory & Marketplace	Give each batch a unique ID. Save the waste type, starting weight, date and source.Record packaged compost. List finished batches, find buyers and track orders or reuse within the society.
Temperature MonitoringImpact & User Access	Show waste processed, compost produced and compost reused or sold. Give each user access only to their allowed tasks.



DOMAIN
EDTECH

Agentic Placement Intelligence Platform

PS DEFINITION:

Colleges + T&P depts run placement on static PDF resumes, spreadsheet eligibility filters, reactive training. Result: unverified skill claims, uncoordinated drives, training late (after hiring season start), lower placement rate. Problem = match, not volume. India Skills Report 2026: graduate employability 56.35%, CS/IT ~78–80%, polytechnic 32.92%. Employers still can't find talent (AI skills hardest). Need one multi-tenant platform: student evidence → verified skill profile → gap-bridging → automated hiring workflow. Explainable, auditable, privacy-safe. Scale single-college pilot → university → multi-college.

SHORT DESCRIPTION:

Multi-tenant platform. Turns student evidence (repos, projects, assessments) into Verified Skill Graph. AI-curated Gap-Bridging Engine closes skill gaps before drives. Placement Orchestrator automates eligibility, drives, shortlists, interviews, offers. Tenant = college, data isolated. Skill never "verified" without evidence. Explainable, auditable outcomes at scale.

CORE FEATURES:

- Evidence-Based Competency Profiling: Ingest repos (with consent) + other evidence. Map to standard taxonomy. Each skill has evidence trail + state: Claimed, Observed, Evidence-linked, Verified, Expired.
- Skill-Gap Analysis + AI Learning Paths: Compare verified profile vs live industry demand + target role. Personal paths + practice tasks. Close gaps pre-recruitment.
- Automated Campus Hiring Workflow: Declarative versioned eligibility rules replace spreadsheets. Coordinate drive scheduling, shortlisting, assessments, interviews, offers (T&P + recruiters).
- Governance, Explainability, Multi-Tenant Trust Layer: Per-college isolation, consent mgmt, explainable scoring, human oversight of AI decisions, fairness monitoring, tamper-evident audit trail.

BUSINESS REQUIREMENTS:

QUASAR 5.0

Core Competency Profiling & Verification

- Student-owned dynamic profile: Profile part of student record. Updated from evidence. Independent of resume file. Resumes + recruiter views generated on demand.
- Evidence-gated verification: Never show verified w/o evidence. Each skill shows state, linked evidence (commits, files, test results, assessments, mentor attestations), confidence, expiry/re-verify date.
- Consent-based repo ingestion: Explicit, revocable consent. Read-only, least-privilege access. Support non-repo evidence (lab work, competitions, certs) → no student disadvantaged.
- Standards-aligned taxonomy: Map to ESCO, O*NET, NSQF. Support portable verifiable credentials (Open Badges 3.0, W3C VC).

Skill-Gap Bridging & Training Intelligence

- Industry-demand alignment: Ingest recruiter JDs + hiring trends. Live in-demand skill view. Compare vs cohort verified skills.
- Personalised AI learning paths: Target verified gaps for chosen role. Recommend projects + practice tasks producing new verifiable evidence, not just course completions.
- Proactive training planning: Cohort gap heatmaps + recommended interventions pre-season. Measure post-intervention gain in verified skills + shortlisting rates.

Automated Campus Hiring Workflows Declarative Eligibility Engine

- Versioned, configurable rules (academics, backlogs, skills, policy limits). Each decision records rule version + inputs → reproducible, appealable.
- Coordinated drive management: Single calendar + workflow for recruiter onboarding, scheduling, shortlisting, assessments, interviews, offer tracking. Prevent clashes.
- Recruiter talent discovery: Searchable, evidence-backed candidate views by verified skills. Reason codes for every match. No ranking by unexplained score.
- Agentic automation + human oversight: AI agents for ingestion, profiling, gap analysis, eligibility, scheduling. Any action affecting candidacy needs T&P officer review/override. Log every agent action.

Trust, Privacy, Explainability & Compliance

- Explainable scoring: Every score/recommendation shows evidence, contributing factors, confidence. Plain language for students, officers, recruiters.
- Privacy + consent: Comply with India DPDP Act 2023 + Rules 2025. Notice + consent, purpose limitation, retention limits, breach notification, data-principal rights, verifiable parental consent, no behavioural monitoring for under-18.
- Fairness monitoring: Monitor selection/shortlisting across gender, category, branch, college tier. Impact-ratio analysis (four-fifths benchmark). Flag disparities for human review. Follow direction of NYC Local Law 144 + EU AI Act high-risk rules.
- Integrity w/o unreliable AI-detection: Verify via repo-specific walkthroughs, live modification tasks, commit-history coherence, mentor attestation. Never penalise on AI-generated-code detection scores.
- Auditability: Append-only, tamper-evident audit trail. Covers profile changes, verification events, eligibility decisions, agent actions, data access.

QUASAR 5.0

Multi-Tenancy, Scalability & Administration

- Tenant data isolation: Each college = separate tenant. DB-level enforcement (e.g. row-level security). Optional siloed storage for strict universities. Automated cross-tenant leak tests.
- Scalable rollout: Pilot → university → multi-college. Hierarchical roles (university, college, department, officer). Per-tenant config, rate limits, encryption keys.
- Unified reporting: One view of Placement Rate, Verified Skill Coverage, Skill-Gap Closure Rate, Shortlist-to-Offer Ratio, Time-to-Shortlist, Recruiter Engagement Score. No fragmented tools.

Streamlined Student & Recruiter Experience

- Central workspace. Students: verified profile, consent settings, learning paths, drive eligibility. Recruiters: requirements, shortlists, feedback. Transparent communication + structured submission of evidence, applications, offers.

EXPECTED OUTCOME:

- A demonstrable AI-powered, evidence-based placement intelligence platform that combines verified competency profiling, skill-gap analysis, AI-curated learning paths, and automated campus hiring workflows under a trusted multi-tenant governance layer.
- The solution will convert student evidence (repositories, projects, assessments) into a dynamic Verified Skill Profile. No skill is shown as verified without proof. It will compare each student's skills with live industry demand, generate learning paths that create new verifiable evidence, and replace spreadsheet filtering with versioned eligibility rules and one coordinated drive workflow.
- AI agents will assist with profiling, eligibility checking, and scheduling, but any action affecting a student's candidacy will need T&P officer approval. Every score and decision will be explainable and auditable, and each college's data will stay isolated. Officers will see key metrics such as Placement Rate, Verified Skill Coverage, and Skill-Gap Closure Rate in one dashboard.
- The prototype will provide a practical foundation for scaling from a single-college pilot to university-wide and multi-college deployment.
- One college. Verified skill profiles. Explainable decisions. Human-controlled automation. Fair and auditable placement outcomes.

PS :- EDTECH ESSENTIAL REQUIREMENTS

QUASAR 5.0

- **STUDENTS DASHBOARD**

Essential Requirement	Description
Student Skill Profile	Every student has one profile page. It fills itself from GitHub, LeetCode, projects, certificates. Resume is made from this profile with one click. Student never uploads PDF. Example: Rahul links GitHub + LeetCode. System pulls his 200 solved problems, 5 projects, Python certificate. Company wants resume → Rahul clicks once → resume ready.
Safe Code Import	Student gives permission to read GitHub repos. Read-only. Student can withdraw anytime. Other proof (labs, certificates, contests) also accepted.
Skill Proof Check	Each skill gets a tag. No proof = "Claimed". Proof found = "Verified". Proof = code on GitHub, test score, or mentor sign-off. Old proof expires, student must refresh. Example: Rahul writes "Python" → Claimed. System finds Python code in GitHub → Verified. Proof 1 year old → Expired.
Target Companies + Skill Gap	Student sees best-fit target companies. Also sees gap between own skills and each company's needs.
Mock Drive + Personal ReportMock Drive + Personal Report	Student picks target role, takes 3 practice rounds: 1) Aptitude: 30 questions, different set for every student. 2) Machine Test: 2 coding questions, 10 marks each. 3) Interview: questions based on chosen role. After all 3, student gets report: score per round, weak topics, where marks were lost. Example: Rahul picks "Backend Developer". Aptitude 21/30. Machine 12/20. Interview weak on database. Report: "Weak in SQL + problem solving."
Personal Learning Path	Roadmap built from Mock Drive report + skill gap. Tells student what to improve. Tasks and projects that create new proof. Example: Rahul weak in SQL → 2 SQL practice sets + 1 mini project → new proof added → retake mock drive.
My Drives + Eligibility	Student sees upcoming drives, whether eligible, and why. If not eligible, sees which rule failed. Can appeal.
My Privacy + Consent	Student controls data sharing, can withdraw GitHub access, sees why every score was given in simple words.

PS :- EDTECH ESSENTIAL REQUIREMENTS

QUASAR 5.0

- **ADMIN DASHBOARD**

Essential Requirement	Description
Cohort Skill Dashboard + Reports	Heatmap of weak skills across batch, branch, college. Shows placement rate, skill coverage, gap closure, offers, time to shortlist. Reports: Mock Drive scores per round, top weak topics, students needing help, before-vs-after improvement. Download as PDF/Excel. Example: CSE batch report shows 60% weak in SQL → officer books SQL workshop → next report shows 25%.
Industry Skill-Gap Profiler Agent	AI agent scans job descriptions across tech. Finds rising skills. Compares them with what the batch can do. Shows the gap. Example: 40% of new jobs ask "Docker". Only 10% of batch has it → officer sees alert → adds Docker training.
Predictive Placement Analytics Agent	AI agent forecasts batch placement rate. Flags at-risk students 6 months before placement season. Uses skills, mock scores, learning progress. Officer reviews list. Example: "Batch placement likely 68%. 45 students at risk." Officer assigns mentors early.
Mock Drive Setup	Officer manages question bank for aptitude and machine tests. Reviews AI-made interview questions per role. Sees all student reports.
Bulk Mailing	Officer sets invitation mail template once. System sends it to all companies in company database in one click. Replies tracked.
Drive Management + Auto Allocation	One calendar for company visits, shortlists, tests, interviews, offers. No date clashes. System reads company's email, auto-picks slot and allocates drive. Officer confirms.
Eligibility Rules	Officer sets rules once (marks, backlogs, skills). System checks every student. Saves rule version used, so decision can be checked or appealed.
Recruiter Candidate Search	Recruiters (officer-approved access) search by verified skills. Each result shows why student matched. No secret scores.
AI Helps, Human Decides	AI does boring work (profiling, checking, scheduling, making interview questions, predictions). Officer must approve anything that affects student's chance. No student labelled "at-risk" or "failing" without human check. All AI actions logged.

PS :- EDTECH ESSENTIAL REQUIREMENTS

- **ADMIN DASHBOARD**

Essential Requirement	Description
Fairness Check	System checks selection results for bias across gender, category, branch. Flags problems for officer to review.
Accreditation Reporting Suite	One-click export. Makes NAAC/NIRF-ready placement reports and career outcome stats. No manual Excel work. Example: NAAC visit coming → officer clicks “Export” → report ready with placement numbers, salary data, higher studies, branch-wise outcomes.
Roles + College-Separate Data	Each college’s data kept fully separate. Roles: university, college, department, officer. Follows India’s DPDP law.



DOMAIN
VIBRATION ENERGY HARVESTING

Vibration-Powered Supply for Low-Energy Truck Electronics

PS DEFINITION:

Vibration Energy Harvesting converts mechanical vibrations from a moving truck into electrical energy. The system captures and conditions this energy, stores it, and provides a regulated 3.3 V supply to low-power electronic devices.

SHORT DESCRIPTION:

Truck electronics, such as tracking units and sensors, require a reliable power supply. Mechanical vibrations from the engine, chassis, suspension, and road movement can serve as an alternative energy source. The proposed system converts these vibrations into electrical energy using a suitable harvesting mechanism. The generated energy is conditioned and stored in a supercapacitor or suitable rechargeable battery for later use. The prototype must demonstrate a nominal 3.3 V supply to a defined electronic load drawing less than 20 mA. Teams must measure performance under load and test the system under controlled vibration conditions.

CORE FEATURES:

- Vibration Energy Harvesting: Convert mechanical vibrations into electrical energy using a piezoelectric, electromagnetic, or other suitable mechanism.
- Power Conditioning and Storage: Condition the generated electricity and store usable energy safely using appropriate power management circuitry and a supercapacitor or rechargeable battery.
- Regulated Power Delivery: Provide a nominal 3.3 V output to a defined low-power electronic load, accounting for conversion losses and variations in energy generation.
- Performance Monitoring: Measure generated voltage and current, storage charging, output stability, load current, and runtime. Test the system when vibration changes or stops.

BUSINESS REQUIREMENTS:

- The solution must integrate energy harvesting, power conditioning, storage, and regulated power delivery into a functional and measurable prototype.
- The implementation approach is open to participants, allowing flexibility in component selection, circuit design, and testing methods.
- The prototype must be safe, repeatable, and supported by documented results. An actual truck installation is not mandatory for the minimum viable prototype if an approved laboratory vibration rig is used.

CORE FEATURES AND ESSENTIAL REQUIREMENTS

A) Vibration Input and Energy Harvesting

- Vibration Input: Use a controlled vibration source representative of truck vibrations and document the test conditions.
- Energy Conversion: Demonstrate measurable electrical energy generation from mechanical vibrations.
- Variable Vibration Testing: Compare performance under different vibration frequencies, amplitudes, or test settings.

B) Power Conditioning, Storage and Output

- Power Conditioning: Convert the raw harvested output into usable electrical energy.
- Energy Storage: Demonstrate energy storage using a supercapacitor or suitable rechargeable battery with appropriate protection.
- Regulated Output: Provide a nominal 3.3 V supply and measure its stability under load.
- Load Capability: Demonstrate operation of a defined electronic device drawing less than 20 mA at 3.3 V. If the target is not achieved, report the maximum verified load and explain the limitations.

C) Performance Testing and Validation

- Performance Measurement: Record generated voltage and current, charging behaviour, output voltage, load current, and runtime.
- Repeatability: Repeat at least one test under identical conditions and document variations.
- Ride Through: Measure how long the load remains powered after vibration stops or decreases.
- Energy Monitoring: Optionally display or log system measurements through a monitoring interface.

D) Mechanical Design and Safety

- Mounting: Propose a secure mounting arrangement and, if possible, a compact enclosure suitable for future vehicle testing.
- Safety: Ensure secure mechanical assembly and safe electrical connections. Actual truck testing requires explicit authorization.

- Cost Analysis: Provide a bill of materials and explain the trade offs between cost, size, efficiency, and complexity.

E) Documentation and Operating Limits

- Technical Documentation: Include a system block diagram, circuit overview, component list, test procedure, results, and limitations.
- Operating Limits: Explain how vibration conditions, load demand, conversion losses, and stored energy affect performance.

EXPECTED OUTCOME

A functional prototype that harvests energy from controlled, truck representative vibrations, stores the generated energy, and delivers a measured nominal 3.3 V output to a low-power electronic load.

The submission must include test results, an energy budget, a component list, and a proposed path toward validation on an actual truck

PS :- VIBRATION ENERGY HARVESTING ESSENTIAL REQUIREMENTS

QUASAR 5.0

Essential Requirement	Description
Fairness Vibration-to-Electric Conversion Check	Demonstrate a working mechanism that converts vibrations representative of a moving truck into measurable electrical energy. Teams may use a piezoelectric, electromagnetic, or other suitable harvesting approach.
Usable Power Conditioning	Condition the raw harvested output so it can be safely used by the target electronic load. Rectification, power management, and voltage conversion should be selected according to the harvester's actual output.
Energy Storage	Demonstrate energy entering a supercapacitor or suitable rechargeable battery so stored energy can be used when vibration or instantaneous generation falls. Use suitable charging and protection circuitry.
Regulated 3.3 V Output	Provide a nominal 3.3 V supply to the load and measure the voltage while the load is connected. Report the achieved voltage range and output stability under load.
Target Load Capability	Demonstrate operation of a defined low-energy electronic load drawing less than 20 mA at 3.3 V, or report the highest verified load and explain any gap to the target. State the load current profile and duty cycle.
Measured Performance	Record relevant measurements including harvester voltage/current, storage charging behaviour, regulated output voltage, load current, runtime, vibration/test settings, and relevant losses where measurable.
Repeatable Test	Repeat at least one test under the same stated conditions and report variation or instability. Clearly describe the vibration input and disclose if a bench rig is used instead of an actual truck.
Safe Prototype	Secure the mechanical assembly, use appropriate electrical connections, and handle batteries or supercapacitors safely. Demonstrations must use an approved test rig unless actual-vehicle testing is explicitly authorized.